Gas Absorption

Experiment 2

CHE381: Unit Operations Laboratory II

Tuesday 9 AM

February 17, 2020

Group 1:

James Fell

Diego Martinez

Allison Nordvall

William Rawson

Emily Sobie

Peter Starakiewicz

Commented [PTH1]: 90/100

Executive Summary

The study of the absorption of CO_2 in water was successfully completed for this three-part experiment. In part one, the objective was to study the specific fluid characteristic of flooding point in the absorption column at 4 different water flowrates. The flooding was found to have occurred when the air flowrate was 80 L/min, 80 L/min, 20 L/min, and 20 L/min with respective water flow rates of 1.5 L/min, 3 L/min, 6 L/min, and 8 L/min. The flooding data was found for the Armfield Gas Absorption Column filled with 1-cm Raschig rings that had calculated void fraction of 0.339. In the second part of the experiment, the CO_2 concentration was found to be 0.016 M at equilibrium with a volume fraction of 0.17. The overall mass transfer coefficient was calculated initially to be $3.85*10^{-4}$, and then once equilibrium had been reached the coefficient increased to $1.29*10^{-2}$. The desorption was also measured in the second part of the experiment and CO_2 found to have a concentration of $3.00*10^{-4}$ at equilibrium. For the third part of the experiment, the absorption of CO_2 was measured for a continuous column mode. The volume fraction for this mode was calculated to be 0.16, with an initial overall mass transfer coefficient of $1.28*10^{-4}$. During this part of the experiment, many samples were taken for the continuous column. The mass transfer coefficient for the samples ranged from $1.28\cdot10^{-4}$ L/min to $1.11\cdot10^{-2}$ L/min. However, 5 samples were calculated to have mass transfer coefficients of around $9.72*10^{-3}$, while the other 6 samples had mass transfer coefficients around $1.23*10^{-2}$.

Commented [T2]: 8/10

Commented [T3]: This doesn't look right. Your flooding point should be at lower air flowrate when you increase the liquid flowrate.

Commented [T4]: Unit?

Commented [T5]: Can you explain more about this part? I don't quite understand; why didn't you calculate mass transfer coefficient for the average value of CO2 concentration (your equilibrium CO2 concentration)

Introduction

Commented [T6]: 10/10

The separation of chemicals is a critical skillset for a chemical engineer. The techniques for separation can be split into 4 major sections: distillation, absorption, extraction, and stripping. This experiment will explore absorption, a unit operation in which one or more gas components are removed by being taken up (absorbed) in a nonvolatile liquid (solvent) [1]. Many process applications depend on absorption for product quality and purity, but it is also implemented to aid in pollution control. One example of absorption is the removal of CO_2 from natural and flue gasses so that it is not added to the atmosphere, helping combat global warming. Another example is the removal of butane and pentane from a refinery gas using a heavy oil [1]. Both of these examples showcase the two mechanisms in which absorption can occur: physical and chemical. Physical absorption requires separation based on the different solubilities of the gases in a specific solvent. Chemical absorption uses a reaction between the gas and the solvent. Therefore, in both chemical and physical absorption the absorbing liquid will be chosen based on its selectivity for the desired components chosen to be removed from the gas.

Overall, the emphasis of this experiment is to study the absorption of CO_2 into water. There are three specific objectives in this experiment that will be studied. The first objective is to study the fluid characteristics in a gas absorption column, such as the flooding point of the column. The second objective is to verify the gas absorption characteristics, such as the composition of the gas sample. The final objective is to determine the overall mass transfer coefficient in the removal of CO_2 with water.

To accomplish these objectives, the Armfield Gas Absorption Column (UOP7) packed with 1-cm Raschig rings column was used. The column inside diameter measured at 8 cm and the rings inside diameter measured at 0.8 cm. Deionized water was used as the nonvolatile solvent and the gas being absorbed was CO_2. The water was filled in the reservoir tank and pumped throughout the column. Control valves and flow meters (specifically a rotameter) were used to control the system, according to the specific conditions determined in the experiment. Regulators and flow meters were also used to control the CO_2/air supply. Additionally, a manometer was attached to the apparatus to determine the pressure drop within the column and a Hempel apparatus was used to determine the volume fraction of CO_2 in air leaving the column. It is also important to note that the column was set up with a counter current flow between the gas and liquid.

Theory

Commented [T7]: 10/10

The theory behind gas absorption is that a soluble vapor is mixed with an inert gas and can be absorbed by a liquid. In our case, we absorbed CO_2 out of the air with liquid water. The most common apparatus in gas absorption is a packed column. The typical configuration for a packed column is a cylindrical column; a gas inlet port at the bottom of the column; a liquid inlet port at the top of the column; gas and liquid outlets at the top and bottom of the column, respectively; and tower packing made from inert solid shapes [2]. The solute-containing gas (also known as rich gas) travels through the packing material upward towards the top of the column, which is countercurrent to liquid flowing towards the bottom of the column. The purpose of the packing material is to provide a large surface area for the gas and liquid to come in contact with each other to facilitate mass transfer. The gas that leaves the top of the column will have a lower concentration of the solute because the solute was absorbed by the liquid (called lean gas). The liquid that flows out the bottom of the column will have a higher concentration of the solute because it absorbed the solute out of the gas stream. The packing material of the column is an inert material and are usually shaped irregularly or hollow so that they can interlock. Typical void fractions of the packing materials in the column are between 60-90%.

An important consideration in the operation of the gas absorption column is to understand at what gas flow rate the column will flood. Flooding is when the whole column fills with liquid and is caused by the increased friction between the gas and liquid phases. It is desired to approach this flooding velocity as close as possible because just before flooding, the maximum contact area between the liquid and gas phase is achieved. The flooding velocity is strongly correlated to the size/shape of the packing material and the liquid mass velocity [2]. An empirical correlation has been developed for the limiting pressure drop for column flooding:

$$\Delta P_{flood} = 0.115 * F_p^{\ 0.7} \qquad (1)$$

Where P_{flood} is the pressure drop at flooding in 2 in. H_2O/ft of packing and F_p is the dimensionless packing factor (F_p = 580 for Raschig Rings used in our experiment) [2].

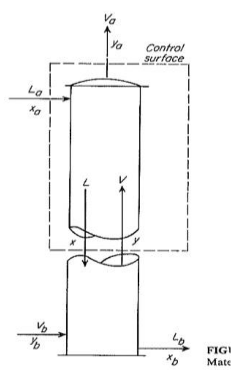


Figure 1: Diagram of the packed column that will be used for the material balance. *Adapted from Unit Operations of Chemical Engineering* [2].

The material balance of an arbitrary control volume around the gas absorption column is given below (See **Figure 1**, control volume is the dashed line).

Total Material: $$L_a + V = L + V_a \quad (2)$$

Species A: $$L_a * x_a + V * y = L * x + V_a * y_a \quad (3)$$

The overall material balance is given below:

Total Material: $$L_a + V_b = L_b + V_a \quad (4)$$

Species A: $$L_a * x_a + V_b * V_b = L_b * L_b + V_a * y_a \quad (5)$$

Where V is the flow rate of the gas stream, L is the flow rate of the liquid stream, x is the composition of the liquid stream, and y is the composition of the gas stream. Equation 3 can be rearranged to find the relationship between x and y and any point in the column; the following equation is called the operating-line equation [2].

Operating Line Equation: $$y = \frac{L}{V} * x + \frac{V_a * y_a - L_a * x_a}{V} \quad (6)$$

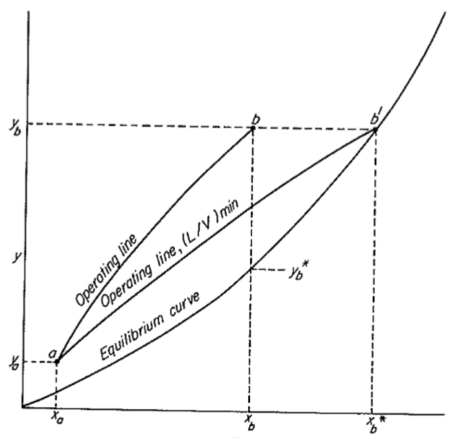


Figure 2: Plot of the operating line and the equilibrium curve. The operating lines are curved because as the solute gets absorbed in the liquid, the gas flow rate decreases through the column while the liquid flow rate increases. *Adapted from Unit Operations of Chemical Engineering* [2].

The determination of how many moles of the solute has been transferred in to the liquid can be calculated using the following correlation:

$$N = ka\Delta C \qquad (7)$$

Where *N* is the moles of the solute transferred, *ka* is the mass transfer coefficient multiplied by the interfacial area, and ΔC is the driving force which can be the difference in partial pressure, concentration, or mole fraction. The number of moles transferred of the solute into liquid is therefore strongly dependent on the contact area between the liquid and gas phase. The mass transfer coefficient is also dependent on the temperature, pressure of the column, and the concentration of the solute in the gas and liquid stream.

Results

The goal of this experiment was to determine fluid flow characteristics, such as the flooding point, in a packed bed absorber. This experiment also had a second objective to determine the gas absorption characteristics and the overall mass transfer coefficient in the removal of CO_2 from an air stream with water. During each part of the experiment, different variables were changed, such as the air flowrate, water flowrate, and the CO_2 flowrate, to obtain the desired results. Before starting the experiment it was important to calculate how much space the raschig rings take up. The void fraction in the column due to the rings was determined to be 0.339.

In part 1 of the experiment, the flooding point of the packed bed column was measured. The flooding point of the column was achieved by adjusting the air flow rate and water flow rate in the column. It could then be determined that the flooding point had been reached when there was a significant change in pressure drop. In order to find the flooding point of the column, 5 trials were completed at different water flow rates. The water flow rates used were 0, 1.5, 3, 6, and 8 L/min. For each trial, the air flow rate started at 20 L/min and was increased by 20 L/min increments until flooding occurred. No flooding occurred when the water flowrate was 0 L/min. Flooding was achieved when the air flowrate was at 80 L/min, 80 L/min, 20 L/min, and 20 L/min when the water flow rate was set at 1.5 L/min, 3 L/min, 6 L/min, and 8 L/min respectively. Figure 3 below shows the column pressure drop (Pa) vs. the gas flowrate (L/min), and the flooding point is marked for each trial. Figure 4 shows the log of the pressure drop vs. the log of the gas flowrate.

Commented [T8]: 12/15

Commented [T9]: So your flooding point is correct for 3L/min water flowrate. However, the 1.5L/min water flowrate should have flooding point around 100L/min air flowrate. Also, at 8L/min, your column is already flooded at the lowest air flowrate so that liquid flowrate doesn't have a flooding point.

Commented [T10]: Explain what's the purpose of this figure? If you put it in your report, you have to explain it.

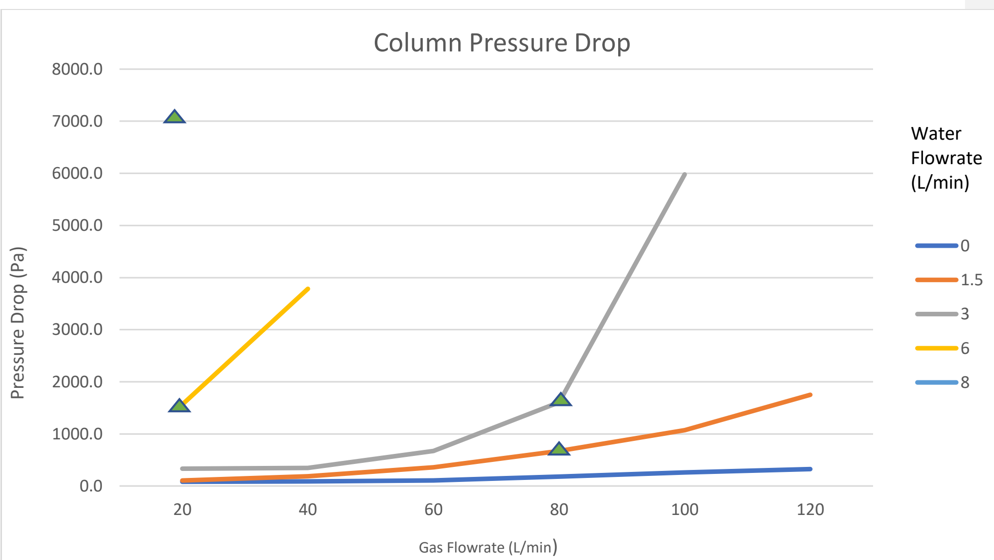


Figure 3: *Column Pressure Drop*

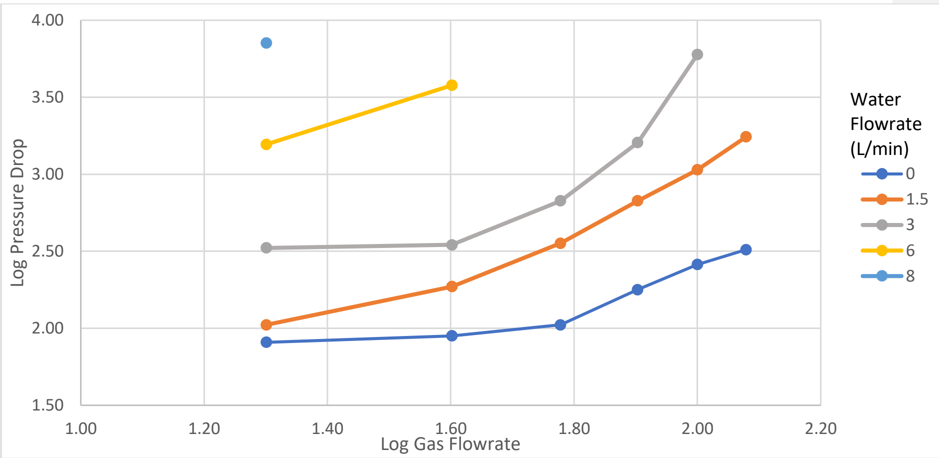


Figure 4: Log of the pressure drop vs. Log of the gas flowrate

In Part 2A, the CO_2 was brought to equilibrium with the water, and the CO_2 concentration was measured. This was done by taking an initial reading of the CO_2 concentration in an initial sample, and then adjusting the water flowrate, the air flowrate, the pressure regulating valve, and the CO_2 flowrate and then letting the system run for 10 minutes. Samples were taken in 50 mL increments and the CO_2 concentration in each sample was measured. The values of CO_2 in each sample is shown in Figure 5 below. The volume fraction of the CO_2 in the air leaving the column after the dissolved CO_2 levels were constant was determined to be 0.17. The mass transfer coefficient is initially at $3.85*10^{-4}$ L/min After the system was allowed to operate for 10 minutes, the mass transfer coefficients ranged at approximately $1.41\cdot10^{-2}$ L/min – $3.85\cdot10^{-4}$ L/min. Figure 6 below shows the CO_2 concentration change across the column for each sample.

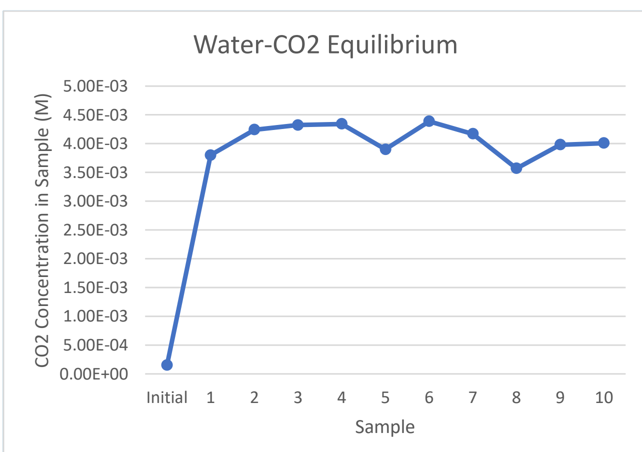


Figure 5: *Water/CO_2 Equilibrium*

Commented [T11]: Plot this with the x-axis as your time, not your sample number. That will show you how long it takes for you system to reach steady state.

Commented [ZA12]: X-axis OK for this because of gap between initial and 1st sample when I was discussing the Hempl apparatus

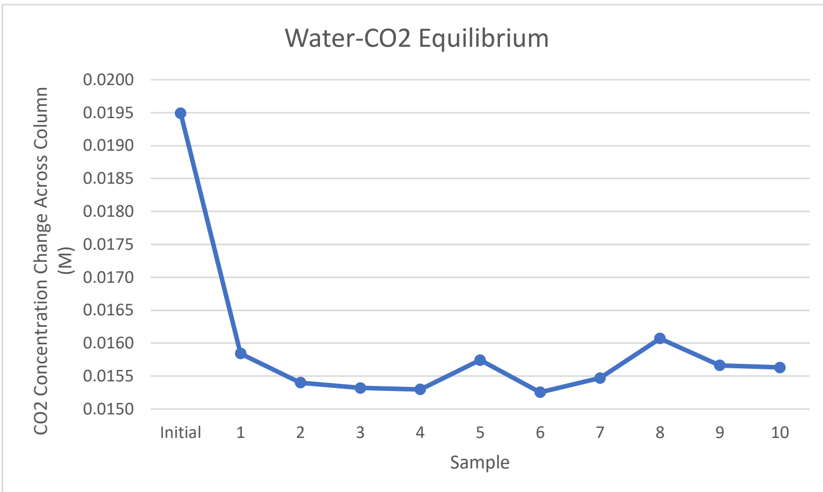


Figure 6: CO_2 concentration change across the column while water and CO_2 are at equilibrium

Commented [T13]: See previous comment about the x-axis

Commented [ZA14]: Same comment as above

In part 2B, the desorption of CO_2 was measured. This was achieved by shutting off the CO_2 flow, decreasing the liquid flow, and opening the drain valve. Samples were taken in 1 minute intervals, and they were analyzed. The CO_2 concentration was determined in each sample and is shown below in Figure 7. The CO_2 concentration in the initial sample was $7.78*10^{-4}$ M. The CO_2 concentration for the samples ranged from $2.00\cdot10^{-3}$ M to $7.78\cdot10^{-4}$ M.

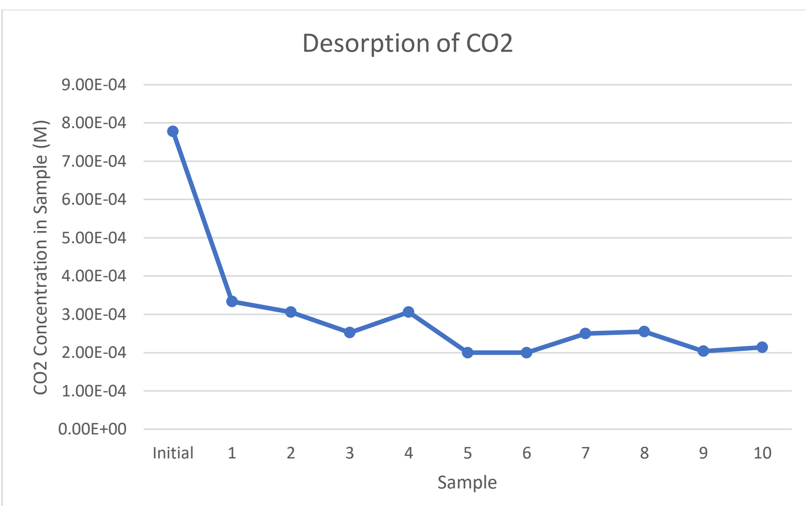


Figure 7: *Desorption of CO_2*

In part 3, the absorption of CO_2 was measured in a continuous column mode. This was achieved by collecting an initial sample and measuring the CO_2 concentration. The air flowrate and the pressure regulating valves were set to 20 L/min and 20 psig respectively. Samples were immediately taken at 30 second intervals for two minutes, and then spaced out to 1 minute intervals until a steady state concentration was reached. The CO_2 concentration in each sample was determined and is shown in Figure 8 below. The volume fraction for this part was determined to be 0.16. The mass transfer coefficient for the initial reading was $1.28 \cdot 10^{-4}$ L/min. The mass transfer coefficient for the samples ranged from $1.28 \cdot 10^{-4}$ L/min to $1.32 \cdot 10^{-2}$ L/min. Half of the samples, 5 to be exact, had a mass transfer coefficients around $9.72 \cdot 10^{-3}$ L/min, while the other 6 samples had a mass transfer coefficients around $1.23 \cdot 10^{-2}$ L/min. Figure 9 below shows the CO_2 concentration across the column while under a continuous absorption of CO_2.

Commented [T15]: Just calculate the mass transfer coefficient for the value of Co2 when the system reaches equilibrium, which is about 1.3E^-2

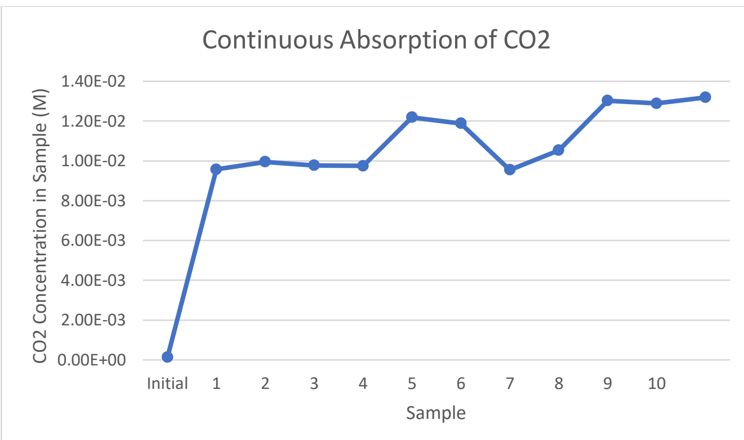


Figure 8: *Continuous Absorption of CO_2*

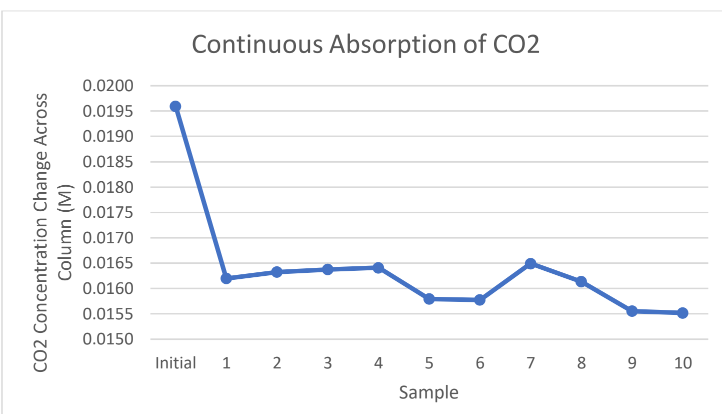


Figure 9: CO_2 concentration change across the column while under a continuous absorption

Discussion

Commented [T16]: 11/15

Very confusing discussion section. Re-attach many Missing some key discussion here. You should compare the mass transfer coefficient between part 2a and 3 and also provide explanation why there is a difference (or no difference) Part 3 was not mentioned in this section. Neither was the desorption of CO2

Good flooding section though.

Packed beds are largely used by Chemical Engineers to make contact between liquids and gas. When preforming this task, there are several characteristics that need to be taken into account. These characteristics include pressure drop and heat transfer between the bed and column wall. In order to optimize the contact between the gas and liquid it is important to choose the proper gas and liquid flow rate to optimize the contact between the two.

In order to accurately discuss the results, it is important to fully understand the three phases that are present in a gas-liquid packed-bed absorption column.

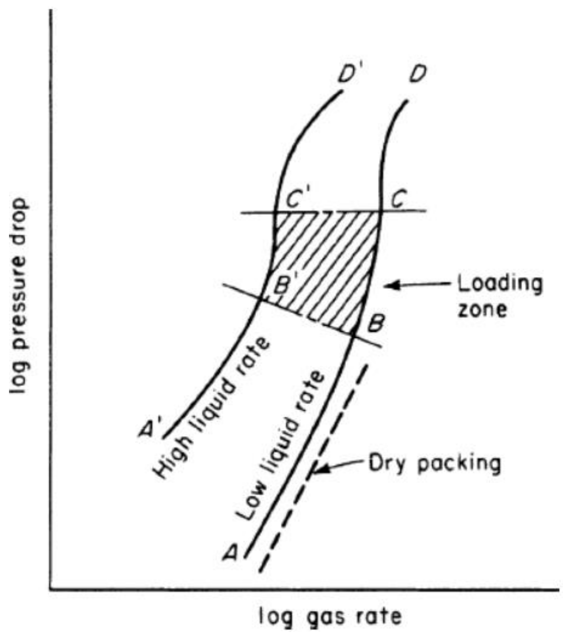


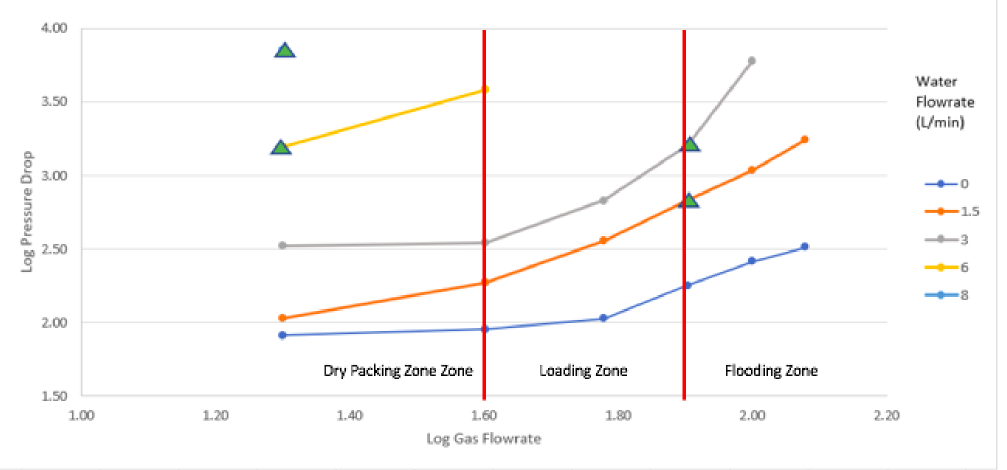
Figure 10. Perry's Chemical Engineers' Handbook of different phase in absorption columns [1]

In order to understand the impact that flooding plays on gas absorption in the column it is important to understand the different phases in the absorption column that are shown in figure 10. The first region of the column is the dry packing zone where liquid does not occupy the void space of the bed. [1] If liquid does not occupy the bed, then the pressure drop within the system is small. Due to the pressure drop being small, the column does not have good efficiency of absorption due to the fact that the contact area for gas and water is minimal. When the bed starts to become filled with water but hasn't started flooding, the loading zone phase is reached. This is when liquid occupies the void area of the bed. As a result, gas-liquid contact is maximized, which leads to a higher gas absorption. This draws the conclusion that the absorption of CO_2will increase when the column becomes closer and closer to flooding. It is also observed that when the water starts to flood the column, a pressure drop is observed. The final stage is the flooding region. This is when the bed becomes overfilled with liquid and the column begins to flood. A rapid increase in pressure drop can be observed when flooding is taking place within the column. Another sign the flooding is taking place is that the column is fully filled with water and there is an excess amount of water above the Raschig rings. Although one might think that the gas absorption efficiency is highest when flooding, the efficiency of the column is reduced due to the poor mass transfer within the system.

The first part of the experiment involves measuring the pressure drop across the column at different water flow rates. This is usually done in industry in order to allow for sizing of pumps. The initial testing of the system is also used to observe and calculate any heat transfer that occurs within the bed and the wall.

In part one of the experiment the pressure drop was measured at different flow rates of water. This was important due to the fact that if the flow rate of gas is much higher than that of the liquid, there will be gas that is spilling out of the top of the bed which leads to poor contact between the liquid and gas. It was also discussed above that if the flow rate of liquid is not high enough then poor contact is made with the gas as well.

Figure 11. Log (Pressure Drop) Vs Log(Gas Flowrate)



The dry packing, loading, and flooding zones are shown in figure 11 and can be used in order to determine the best flow rates for both liquid and gas. Using Figure 11, the best gas and liquid flow rate can be chose. The flowrates determined to have the best absorption were 1.5-3 L/min for the flow rate of water and 1.60-1.90 in the logarithm gas flowrate.

The mass transfer coefficient is a flow dependent variable that varies based on the change in concentration of CO_2 that is diffused into the water. Mass Transfer coefficients affect almost all industrial processes but affect each process in different degrees. For absorption, the mass transfer coefficient determines the height of the packed tower and is in the units of volumetric flow (L/min) [4].

When looking at the mass transfer coefficient during loading, there is a dramatic increase in the liquid-gas area which increases the mass transfer coefficient. This agrees with the idea that absorption is best in the loading zone which also means that the mass transfer coefficient is highest in the loading zone. This can

Commented [T17]: Figure caption is below the figure

Commented [T18]: Chosen*

Commented [T19]: Convert this log value to L/min air flowrate be seen in part 3 of the experiment but cannot be seen in part 2 due to the delayed start in taking water samples. This experimental error will be further discussed in the error section.

Using the mass transfer coefficients obtained in part three of the experiment, it can be seen that the highest concentration of CO_2 is between samples 5 and 6. The mass transfer coefficients were at 1.22E-02 and 1.19E-02 compared to the starting mass transfer coefficient of 1.28E-04. This means that the loading zone was reached when samples 5 and 6 were taken. Samples 9,10, and 11 seem to have a higher CO_2 concentration but this must be a mistake due to this being the flooding zone, which would lead to gas leakage from the top of the column. Mass transfer coefficient and the concentration of CO_2 within the sample have a direct relationship and will continue to change based on the flow rate of water and gas that is being supplied. Overall, a higher concentration of CO_2in the sample results in a higher mass transfer coefficient.

Commented [T20]: How can your column go from loading to flooding? The water and air flowrate are kept constant throughout this part. I don't understand what you're trying to say here.

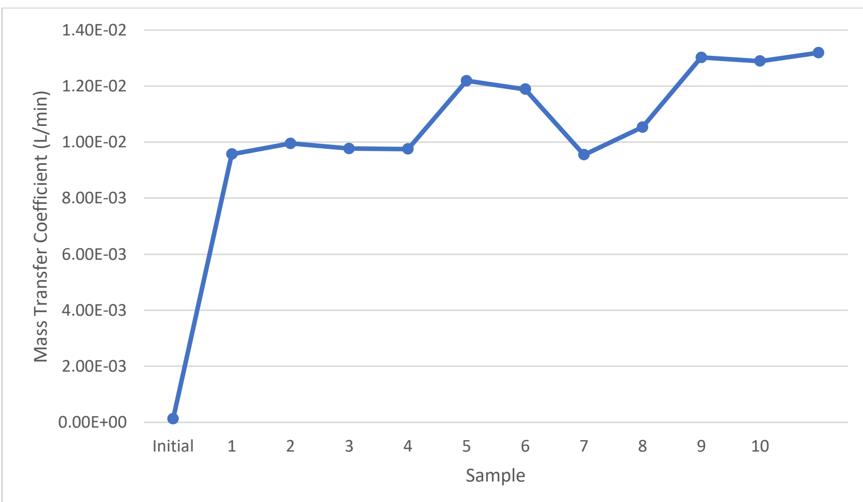


Figure 10: Mass Transfer Coefficients of CO2

Commented [T21]: This looks exactly like the figure for CO2 concentration vs. sample above.

Part 2a of the lab consisted of bringing the water and CO2 components to equilibrium in the column. This means that flowrates and overall interactions of water and CO2 in the column were unchanging over a period of time. In this experiment, this would be observed by having a consistent CO2 concentration change at any point in time. Figure 6 visualized by comparing the change in CO2 concentration entering the column versus the concentration in a sample taken at the end of the column.

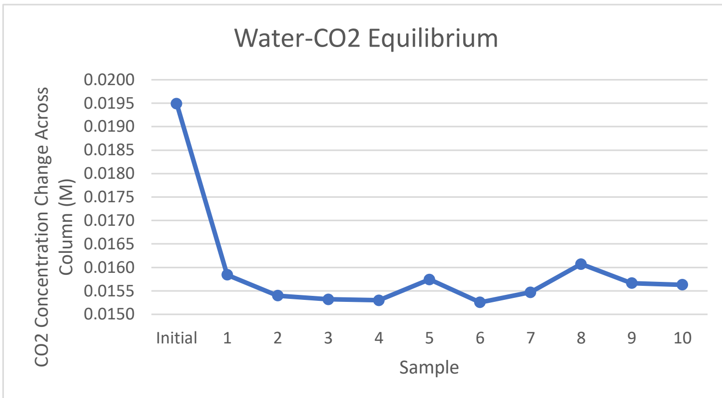


Figure 6: CO_2 Concentration Across Column During Water-CO_2 Equilibrium

It can be seen from the figure that as samples were collected, the CO2 concentration change stayed consistent. Therefore, it can be concluded that water and CO2 had reached equilibrium in the column. In the case of the mass transfer rate and coefficient, these results would indicate that the values for these two properties would be the same at any time after equilibrium was reached.

Absorption and desorption effects of CO2 and water in the column were also measured. Desorption was performed as a follow-up from the equilibrium setup, where no new CO2 was added to the system and instead was only removed. In the opposite fashion, absorption was conducted by having a continuous flow of water through the column. Neither of these setups reached equilibrium as they were both missing a recycled water line. By providing a continuous new source of water or only removing it from the system, equilibrium could not be reached. In desorption it was expected that CO2 would continuously drop in concentration and in absorption concentration would increase. This can be seen in figures 7 and 8;

Commented [T22]:

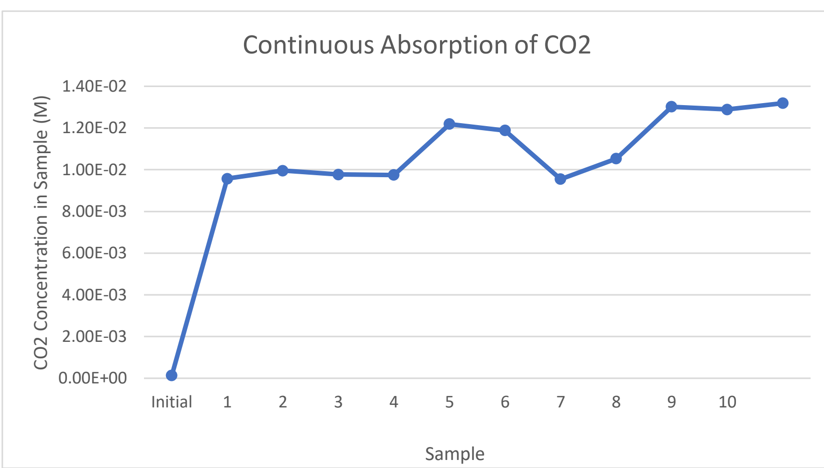


Figure 7: CO_2 Concentration during Desorption

Commented [T23]: I don't understand why you put your figure in the results section to the discussion section. The discussion section is used solely to talk about the results. No need to re-attach any result figure here. Simple mention "figure # from the results section showed that……"
Also, this figure and figure 10 above confused me. They're look exactly the same except the y-axis label.

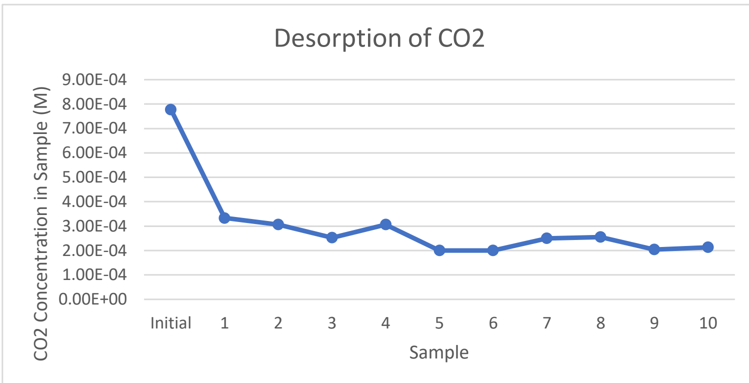


Figure 8: CO_2 Concentration during Absorption

Commented [T24]: You meant desorption?

The figures show that once consistent sampling had begun a slow change in CO2 concentration was observed. In desorption the results were as expected, in which the concentration slowly decreased in each sample. Absorption also went as expected as CO2 would continuously increase in sample concentrations. The overall trends match other similar experiments that were ran on absorption/desorption of CO2 in water. The experimental trends correctly followed an initial sharp jump in measured concentration and then turned into a slow climb as was described in literature [5].

References

Commented [T25]: 9/10
Some of the references aren't cited in the report.

[1] Dr. Don W. Green; Dr. Marylee Z. Southard. Perry's Chemical Engineers' Handbook, 9th Edition (McGraw-Hill Education: New York, Chicago, San Francisco, Athens, London, Madrid, Mexico City, Milan, New Delhi, Singapore, Sydney, Toronto, 2019). https://www-accessengineeringlibrary-com.proxy.cc.uic.edu/content/book/9780071834087

[2] McCabe, Warren Lee, et al. *Unit Operations of Chemical Engineering*. McGraw-Hill, 2005.

[3] Morteza Afkhamipour, and Masoud Mofarahi. "Review on the Mass Transfer Performance of CO 2 Absorption by Amine-Based Solvents in Low- and High-Pressure Absorption Packed Columns." *RSC Advances*, Royal Society of Chemistry, 23 Mar. 2017, pubs.rsc.org/en/content/articlehtml/2017/ra/c7ra01352c.

[4] Cussler. "Absorption." *Calliope*, calliope.dem.uniud.it/CLASS/IMP-CHIM/C9-Cussler.pdf.

[5] Mindaryani, A., et al. "Continuous Absorption of CO2 in Packed Column Using MDEA Solution for Biomethane Preparation." *IOP Conference Series: Materials Science and Engineering*, vol. 162, no. 1, 2016, pp. 1–7, doi:10.1088/1757-899X/162/1/012006.

Appendix I: Data Tabulation

Commented [T26]: 10/10

Table 1: Water - CO_2 Equilibrium Data

Sample #	Initial	1	2	3	4	5	6	7	8	9	10
Sample Volume (mL)	50	50	49.5	48	49.5	50	49	49.5	49	49	48
NaOH Volume (mL)	0.15	3.80	4.20	4.15	4.30	3.90	4.30	4.13	3.50	3.90	3.85
CO2 Sample Concentration (M)	1.50E-04	3.80E-03	4.24E-03	4.32E-03	4.34E-03	3.90E-03	4.39E-03	4.17E-03	3.57E-03	3.98E-03	4.01E-03
CO2 Mass Transfer (mol)	7.50E-06	1.90E-04	2.10E-04	2.08E-04	2.15E-04	1.95E-04	2.15E-04	2.07E-04	1.75E-04	1.95E-04	1.93E-04
Change in CO2 Concentration (M)	0.0195	0.0158	0.0154	0.0153	0.0153	0.0157	0.0153	0.0155	0.0161	0.0157	0.0156
Mass Transfer Coefficient	3.85E-04	1.20E-02	1.36E-02	1.35E-02	1.41E-02	1.24E-02	1.41E-02	1.33E-02	1.09E-02	1.25E-02	1.23E-02
Gas Sample Volume (mL)	6.6										
Syringe Gas Volume (mL)	40										
Volume Fraction	0.17										
CO2 Inlet Concentration (M)	0.0196										
Water Flowrate (L/min)	6										
Air Flowrate (L/min)	20										
CO2 Flowrate (L/min)	5										

Table 2: Continuous Absorption of CO_2

Sample #	Initial	1	2	3	4	5	6	7	8	9	10
Sample Volume (mL)	45	45	49	49.5	49	50	50	50	49	49	46.8
NaOH Volume (mL)	0.7	0.3	0.3	0.25	0.3	0.2	0.2	0.25	0.25	0.2	0.2
CO2 Sample Concentration (M)	7.78E-04	3.33E-04	3.06E-04	2.53E-04	3.06E-04	2.00E-04	2.00E-04	2.50E-04	2.55E-04	2.04E-04	2.14E-04
Water Flowrate (L/min)	1.5										

Table 3: Desorption of CO_2

Sample #	Initial	1	2	3	4	5	6	7	8	9	10	11
Sample Volume (mL)	49	45	49	49	49.5	50	48.5	50	48.5	49.5	48.5	50
NaOH Volume (mL)	0.05	3.1	3.25	3.2	3.2	3.85	3.75	3.15	3.4	4.05	4	4.1
CO2 Sample Concentration (M)	5.10E-05	3.44E-03	3.32E-03	3.27E-03	3.23E-03	3.85E-03	3.87E-03	3.15E-03	3.51E-03	4.09E-03	4.12E-03	4.10E-03
CO2 Mass Transfer (mol)	2.50E-06	1.55E-04	1.63E-04	1.60E-04	1.60E-04	1.93E-04	1.88E-04	1.58E-04	1.70E-04	2.03E-04	2.00E-04	2.05E-04
Change in CO2 Concentration (M)	0.0196	0.0162	0.0163	0.0164	0.0164	0.0158	0.0158	0.0165	0.0161	0.0156	0.0155	0.0155
Mass Transfer Coefficient	1.28E-04	9.57E-03	9.95E-03	9.77E-03	9.75E-03	1.22E-02	1.19E-02	9.55E-03	1.05E-02	1.30E-02	1.29E-02	1.32E-02
Water Temperature (C)	27.3											
Water flowrate (L/min)	1.5											
Air Flowrate (L/min)	20											
CO2 Flowrate (L/min)	5											
Gas Sample Volume (mL)	6.5											
Syringe Gas Volume (mL)	40											
Volume Fraction	0.16											
CO2 Inlet Concentration (M)	0.0196											

Table 4: Packed Bed Pressure Drop – No Water Flow

Water Flowrate (L/min)	Air Flowrate (L/min)	Pressure drop (cm red oil)	Pressure drop (Pa)
0	20	1	81.0
	40	1.1	89.1
	60	1.3	105.3
	80	2.2	178.3
	100	3.2	259.3
	120	4	324.1

Table 5: Packed Bed Pressure Drop – 1.5 L/min Water

Water Flowrate (L/min)	Air Flowrate (L/min)	Pressure drop (cm red oil)	Pressure drop (Pa)
1.5	20	1.3	105.3
	40	2.3	186.4
	60	4.4	356.5
	80	8.3	672.6
	100	13.2	1069.6
	120	21.6	1750.3

Table 6: Packed Bed Pressure Drop – 3 L/min Water

Water Flowrate (L/min)	Air Flowrate (L/min)	Pressure drop (cm red oil)	Pressure drop (Pa)
3	20	4.1	332.2
	40	4.3	348.4
	60	8.3	672.6
	80	19.8	1604.4
	100	73.8	5980.1

Table 7: Packed Bed Pressure Drop – 6 L/min Water

Water Flowrate (L/min)	Air Flowrate (L/min)	Pressue drop (cm red oil)	Pressure drop (Pa)
6	20	19.3	1563.9
	40	46.7	3784.1

Table 8: Packed Bed Pressure Drop – 8 L/min Water

Water Flowrate (L/min)	Air Flowrate (L/min)	Pressue drop (cm red oil)	Pressure drop (Pa)
8	20	88	7130.6928

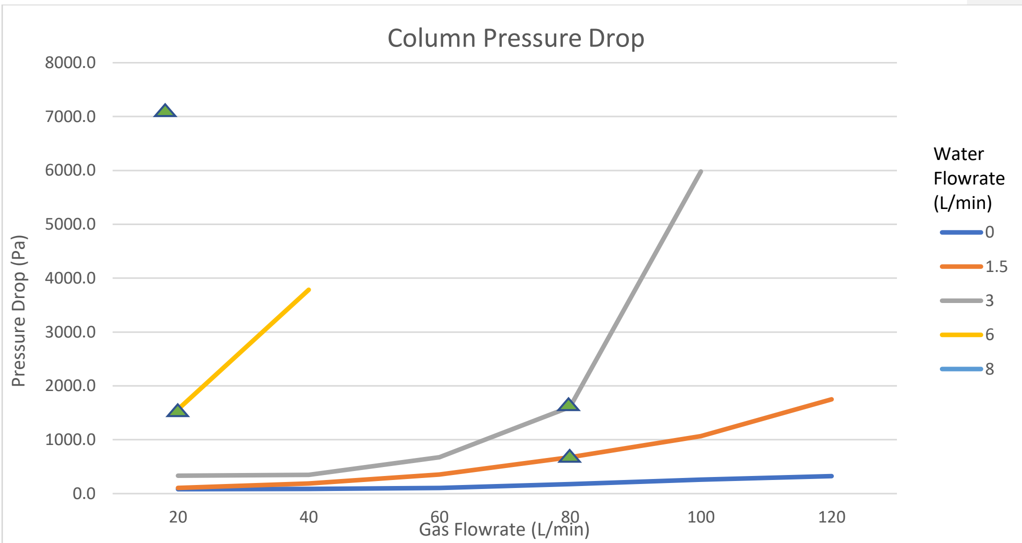


Figure 3: Column Pressure Drop

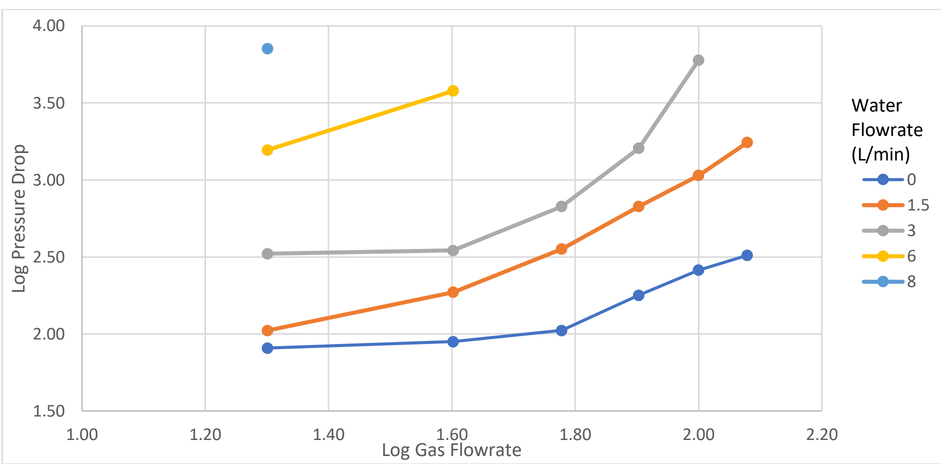


Figure 4: Log of the Gas Flow Rate Vs the Log of the Pressure Drop

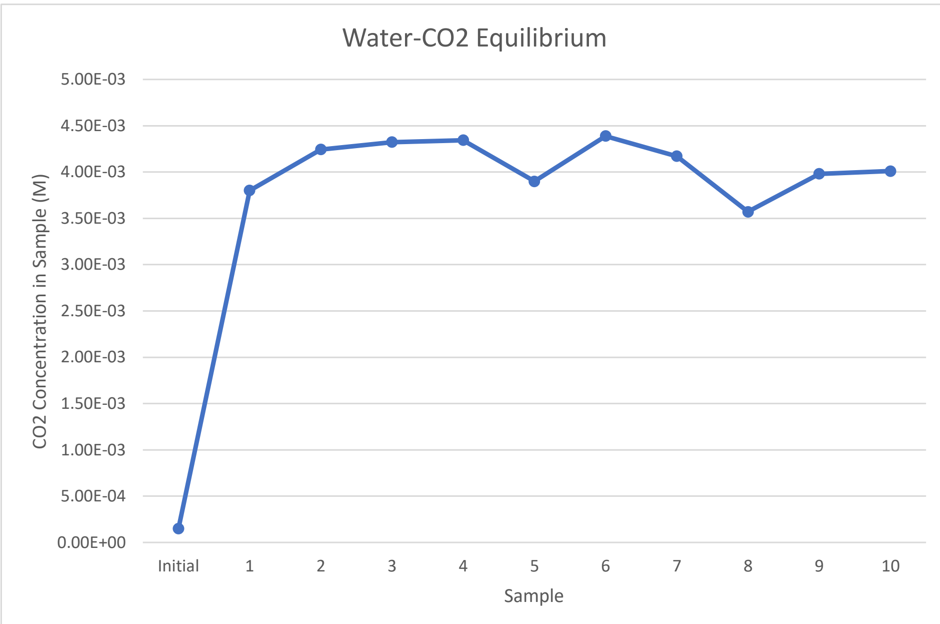


Figure 5: CO_2 Volume during Water-CO_2 Equilibrium

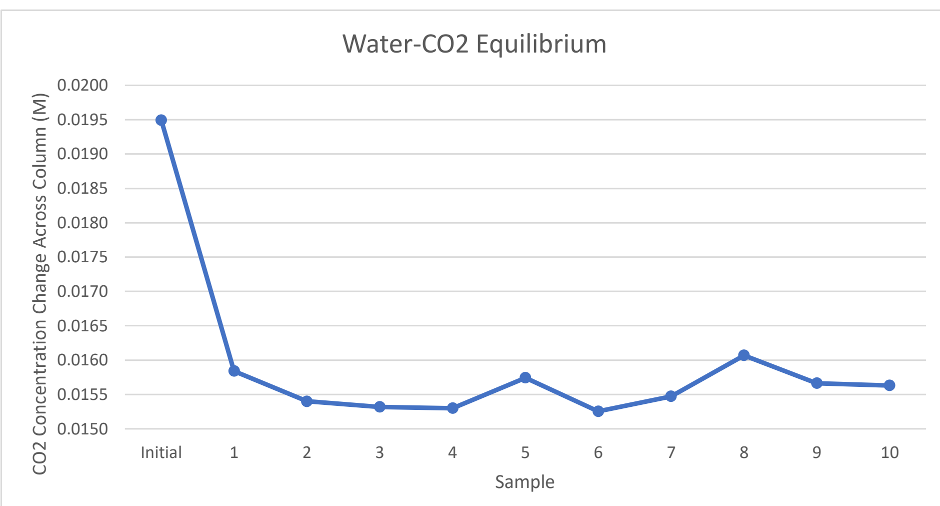


Figure 6: CO_2 Concentration Across Column During Water-CO_2 Equilibrium

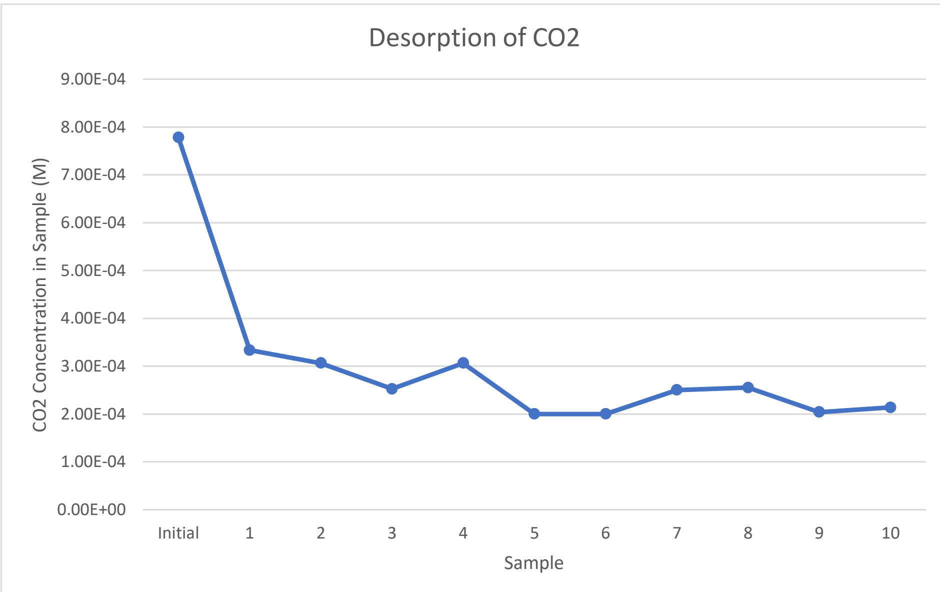


Figure 7: CO_2 Volume during Desorption

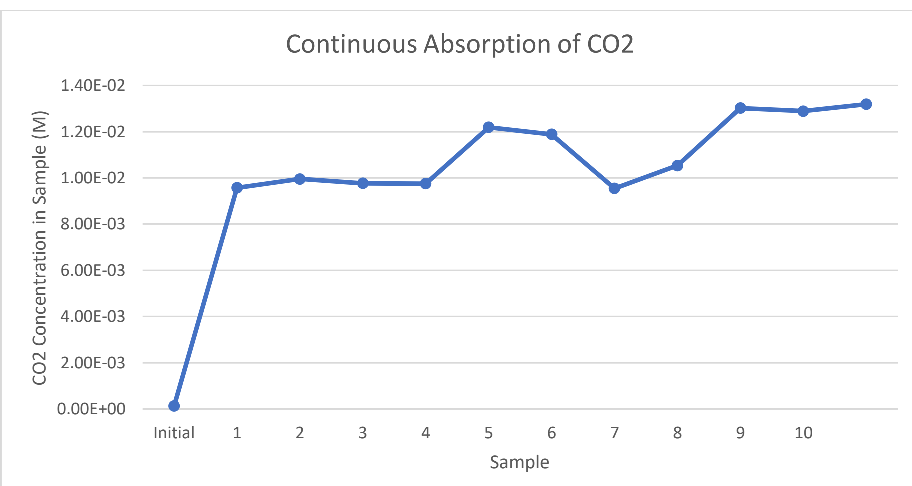


Figure 8: CO_2 Volume during Continuous Absorption

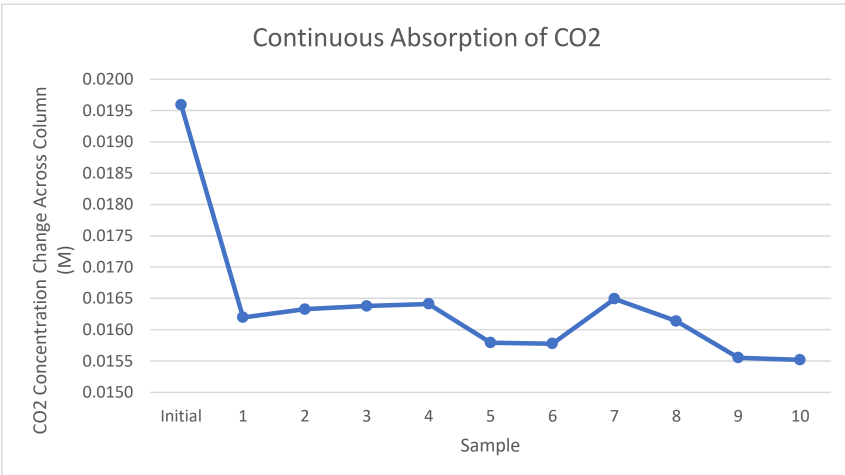


Figure 9: CO_2 Concentration Across Column During Continuous Absorption of CO_2

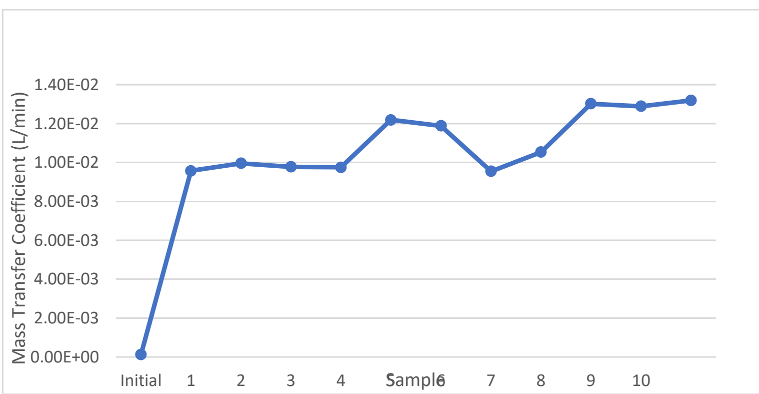


Figure 10: Mass Transfer Coefficient of CO2

Appendix II: Error Analysis

Commented [T27]: 10/10

Throughout the experiment there are several things that could have gone wrong that changed the results that were achieved. Some of the places where there was room for error included during titration. If too much NaOH was added to the water solution, then this would lead to a higher free CO2 sample than was actually taken.

There was also an issue when initially starting the experiment. There was a section where carbon dioxide was being run into the water and was supposed to be left for 10 minutes before taking data points. The system ran for around 15 minutes due to the Hempel column being explained by Dr. Zdunek. This can be seen in the equilibrium graph. The graph starts at the initial then shoots up to become fairly steady. There were not any data points taken leading up to the equilibrium point which means that CO2 equilibrium was already reached when titrations started to be completed.

Finally, there was room for error between transferring the sample to an Erlenmeyer flask and then waiting for titration. There were a few samples that were not covered all of the way when waiting to be titrated. This leaves room for the carbon dioxide to leave the water. This will change the amount of NaOH needed and would then change the Free CO2 sample data point.

Appendix III: Sample Calculations

Commented [T28]: 10/10

Void Fraction

$$\in = \frac{V_{void}}{V_{apparent}} = \frac{205\ mL}{605\ mL} = \ 0.339$$

V_{void} can be defined as the volume of the voids within the column

$V_{apparent}$ can be defined as the total volume of the packed bed

Hempel Apparatus

$$V_{frac} = \frac{V_{Sample}}{V_{Syringe}} = \frac{6.6\ mL}{40\ mL} = 0.17$$

V_{frac} can be defined as the volume fraction of CO_2 in the air sample

V_{Sample} can be defined as the volume of CO_2 dissolved in the NaOH

$V_{Syringe}$ can be defined as the total volume of air sampled in the syringe

Initial CO_2 Concentration

$$C_{i,CO_2} = \frac{\dot{V}_{CO_2}}{\dot{V}_{Gas}} * \frac{P_{CO_2}}{RT}$$

$$C_{i,CO_2} = \frac{5\ L/min}{(20+5)\ L/min} * \frac{2.381\ atm}{\left(0.08206\ \frac{L\ atm}{mol\ K}\right) * (295.45\ K)}$$

$$C_{i,CO_2} = 0.0196\ M$$

C_{i,CO_2} can be defined as the concentration of CO_2 entering the column

$\dot{V}_{CO_2}$ can be defined as the volumetric flow rate of CO_2

$\dot{V}_{Gas}$ can be defined as the total volumetric flow rate of CO_2 and air

P_{CO_2} can be defined as the pressure of CO_2

R is the ideal gas constant

T is the temperature of CO_2

CO_2 Concentration

$$V_B * C_B = V_{Sample} * C_{CO_2}$$

$$C_{CO_2} = \frac{0.15\ mL * 0.05\ M}{50\ mL}$$

$$C_{CO_2} = 1.5\ x\ 10^{-4}\ M$$

V_B can be defined as the volume of based used

C_B can be defined as the concentration of base

V_{Sample} can be defined as the sample volume

C_{CO_2} can be defined as the CO_2 concentration in the sample

Mass Transfer Rate

$$C_{CO_2} = \frac{n_{CO_2}}{V_{Sample}}$$

$$\dot{n}_{CO_2} = C_{CO_2} * V_{Sample} = 1.5\ x\ 10^{-4}\ \frac{mol}{L} * 50\ mL/min * \frac{1\ L}{1000\ mL}$$

$$n_{CO_2} = 7.5\ x\ 10^{-6}\ mol/min$$

$\dot{n}_{CO_2}$is the mass transfer rate, in units per minute as a sample was taken every minute

Mass Transfer Coefficient

$$\Delta CO_2 = C_{i,CO_2} - C_{CO_2} = 0.0196\ M - 1.5\ x\ 10^{-4}\ M$$

$$\Delta CO_2 = 0.0195\ M$$

$$\dot{n}_{CO_2} = ka * \Delta CO_2$$

$$Ka = \frac{\dot{n_{CO_2}}}{\Delta CO_2} = \frac{7.5\ x\ 10^{-6}\ mol/min}{0.0195\ M}$$

$$\text{Ka} = 3.85\ x\ 10^{-4}\,\frac{L}{min}$$

ΔCO_2 can be defined as the change in CO2 concentration over the column, the driving force

Ka can be defined as the mass transfer coefficient

Appendix IV: Individual Contributions

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6.15	All
Pre-Lab Editing	1	All
Experimental Objective (Prelab)	0.5	Emily
Apparatus (prelab)	0.25	Peter/Allison
Materials & Supplies (prelab)	0.5	Peter/Allison
Procedure (prelab)	0.75	William / James
Project Safety Evaluation (prelab)	0.5	Emily / Diego
Final Lab Editing	1.5	All
Executive Summary (final lab)	1	William / James
Introduction (Final lab)	1	William / James
Theory (Final Lab)	3	William / James
Results (final lab)	4	Emily
Discussion (final lab)	4	Allison / Peter/ Diego
References (final and/or prelab)	0.5	All
Data Tabulation/Graphs (final)	3	Allison / Peter
Error Analysis (final lab)	2	Allison / Peter
Sample Calculations (final Lab)	2	Allison / Peter
Power Point Presentation	2.5	All
Total	33.65	-